MH3600 Topology and Manifolds

Solutions of Tutorial #4

Topics covered:

- Continuity

Problem 1

Prove that for functions $f : \mathbb{R} \to \mathbb{R}$, the $\epsilon - \delta$ definition of continuity implies the open set definition.

Solution: Recall that that $f : \mathbb{R} \to \mathbb{R}$ is continuous at a point $x \in \mathbb{R}$ if, for every real $\epsilon > 0$, there is a real $\delta > 0$ such that $|f(y) - f(x)| < \epsilon$ for every real y where $|y - x| < \delta$. We say that f itself is continuous if it is continuous at every $p \in \mathbb{R}$.

Suppose that $f : \mathbb{R} \to \mathbb{R}$ is continuous by the $\epsilon - \delta$ definition above. We show that this implies the open set definition by showing that f satisfies (4) in Theorem 18.1. So consider any $x \in \mathbb{R}$ and any neighborhood V of $f(x)$. Then of course there is a basis element (c, d) containing $f(x)$ such that $(c, d) \subset V$. Let $\epsilon = \min\,(f(x) - c, d - f(x))$, noting that $\epsilon > 0$ since $c < f(x) < d$. It is then trivial to show that $(f(x) - \epsilon, f(x) + \epsilon) \subset (c, d) \subset V$ and contains x. Then, since f is continuous at x, there is $\delta > 0$ such that $|y - x| < \delta$ implies that $|f(y) - f(x)| < \epsilon$ for any real y. Let $U = (x - \delta, x + \delta)$, which is clearly a neighborhood of x. Now consider any $z \in f(U)$ so that $z = f(y)$ for some $y \in U$. Then we have that $x - \delta < y < x + \delta$ so that clearly $-\delta < y - x < \delta$, from which it follows that $|y - x| < \delta$. We then know that $|z - f(x)| = |f(y) - f(x)| < \epsilon$ since f is continuous. Hence $-\epsilon < z - f(x) < \epsilon$ so that $f(x) - \epsilon < z < f(x) + \epsilon$, and thus $z \in V$ since $(f(x) - \epsilon, f(x) + \epsilon) \subset V$. Since $z \in f(U)$ was arbitrary, this shows that $f(U) \subset V$, which shows that (4) holds for f since x was also arbitrary.

Problem 2

Suppose that $f : X \to Y$ is continuous. If x is a limit point of the subset A of X, is it necessarily true the $f(x)$ is a limit point of $f(A)$?

Solution: This is not necessarily true. As a counterexample consider a constant function $f : X \to Y$ defined by $f(x) = y_0$ for any $x \in X$ and some $y_0 \in Y$. It was shown in Theorem 18.2 part (a) that this is continuous. However, clearly $f(A) = \{y_0\}$ for any subset A of X. So even if x is a limit point of A, no neighborhood of $f(x)$ can intersect $f(A)$ in a point other than $f(x) = y_0$ since y_0 is the only point in $f(A)$! Therefore $f(x)$ is not a limit point of $f(A)$.

Problem 3

Let X and X' denote a single set in two topologies $\mathcal{T}$ and $\mathcal{T}'$, respectively. Let $i : X' \to X$ be the identity function.

(a) Show that i is continuous $\Leftrightarrow \mathcal{T}'$ is finer than $\mathcal{T}$.

(b) Show that i is a homeomorphism $\Leftrightarrow \mathcal{T}' = \mathcal{T}$.

Solution: **(a)** First note that clearly the inverse of the identity function is itself with the domain and image reversed, and that for any subset $A \subset X = X'$ we have $i(A) = i^{-1}(A) = A$. ($\Rightarrow$) Suppose that i is continuous and consider any open set $U \in \mathcal{T}$. Then we have that $i^{-1}(U) = U$ is open in $\mathcal{T}'$ since i is continuous. Since U was arbitrary, this shows that $\mathcal{T} \subset \mathcal{T}'$ so that $\mathcal{T}'$ is finer. ($\Leftarrow$) Now suppose that $\mathcal{T}'$ is finer so that $\mathcal{T} \subset \mathcal{T}'$. Consider any open set $U \in \mathcal{T}$ so that also clearly $U \in \mathcal{T}'$, i.e. U is also open in $\mathcal{T}'$. Since $i^{-1}(U) = U$, this shows that i is continuous by the definition of continuity.

(b) Clearly i is a bijection since its domain and image are the same set, and $i^{-1} = i$. We then have that i is a homeomorphism $\Leftrightarrow i$ and i^{-1} are both continuous $\Leftrightarrow \mathcal{T}'$ is finer than $\mathcal{T}$ and $\mathcal{T}$ is finer than $\mathcal{T}'$ (by part (a) applied twice) $\Leftrightarrow \mathcal{T} \subset \mathcal{T}'$ and $\mathcal{T}' \subset \mathcal{T} \Leftrightarrow \mathcal{T}' = \mathcal{T}$, as desired.

Problem 4

Given $x_0 \in X$ and $y_0 \in Y$, show that the maps $f : X \to X \times Y$ and $g : Y \to X \times Y$ defined by

$$f(x) = (x, y_0) \quad \text{and} \quad g(y) = (x_0, y)$$

are imbeddings.

Solution: We only show that f is an imbedding of X in $X \times Y$ as the argument for g is entirely analogous. First, it is easy to see and trivial to formally show that f is injective. The function f can be of course be defined as $f(x) = (f_1(x), f_2(x))$ where $f_1 : X \to X$ is the identity function and $f_2 : X \to Y$ is the constant function that maps every element of X to y_0. Since these have both been proven to be continuous in the text, it follows that f is continuous by Theorem 18.4. Now let f' be the function obtained by restricting the range of f to $f(X) = \{(x, y_0) \mid x \in X\}$. Since f is injective, it follows that f' is a bijection. It follows from Theorem 18.2 part (e) that f' is continuous. Clearly the inverse function f'^{-1} is equal to the projection function π_1 so that $f'^{-1}(x, y) = x$. This was shown to be continuous in the proof of Theorem 18.4. This suffices to show that f' is a homeomorphism, which shows the f is an imbedding of X in $X \times Y$.

Problem 5

Show that the subspace (a, b) of $\mathbb{R}$ is homeomorphic with $(0, 1)$ and the subspace $[a, b]$ of $\mathbb{R}$ is homeomorphic with $[0, 1]$.

Solution: First we show that (a, b) is homeomorphic to $(0, 1)$. First let $X = (a, b)$ and $Y = (0, 1)$, and define the map $f : X \to Y$ by

$$f(x) = \frac{x - a}{b - a}$$

for any $x \in X$, noting that this is defined since $a < b$ so that $b - a > 0$. It is trivial to show that f is a bijection. Now, f is an affine function that could just as well be defined as a map from $\mathbb{R}$ to $\mathbb{R}$, and clearly this would be continuous by basic calculus. It then follows from Theorem 18.2 part (d) that restricting its domain to X means that it is still continuous. We also clearly have from basic algebra that its inverse is the function $f^{-1} : Y \to X$ defined by

$$f^{-1}(y) = a + y(b - a)$$

for $y \in Y$. As this is also linear, it too is continuous by the same argument. This suffices to show that f is a homeomorphism.

The exact same argument shows that $[a, b]$ is homeomorphic to $[0, 1]$ by simply setting $X = [a, b]$ and $Y = [0, 1]$ in the above proof. It is assumed that here again $a < b$ even though the interval $[a, b]$ is valid if $a = b$ and simply becomes $[a, b] = [a, a] = \{a\}$. However, clearly this set cannot be homeomorphic to $[0, 1]$ since it is finite whereas $[0, 1]$ is uncountable.

Problem 6

Find a function $f : \mathbb{R} \to \mathbb{R}$ that is continuous at precisely one point.

Solution: For any real x define

$$f(x) = \begin{cases} 0 & x \in \mathbb{Q} \\ x & x \notin \mathbb{Q} \end{cases}$$

We claim that this is continuous only at $x = 0$. As it is easier to do so, we show this using the $\epsilon - \delta$ definition of continuity, which we know implies the open set definition by Problem 1. First we note that $f(0) = 0$ since 0 is rational. Now consider any $\epsilon > 0$ and let $\delta = \epsilon$. Suppose real y where $|y - 0| = |y| < \delta$. If y is rational then $y = 0$ so that $|f(y) - f(0)| = |0 - 0| = |0| = 0 < \epsilon$. If y is irrational then $|f(y) - f(0)| = |y - 0| = |y| < \delta = \epsilon$ again. Since ϵ was arbitrary this shows that f is continuous at $x = 0$.

Now consider any $x \neq 0$. Let $\epsilon = |x|/2$, noting that $\epsilon > 0$ since $x \neq 0$ so that $|x| > 0$. Now consider any $\delta > 0$. There are two cases:

Case $x \in \mathbb{Q}$. Then $f(x) = 0$ but there is clearly an irrational y close enough to x so that $|y - x| < \min(\epsilon, \delta)$, and hence both $|y - x| < \epsilon$ and $|y - x| < \delta$. We also have that $f(y) = y$. We then have that

$$2\epsilon = |x| = |x - 0| \leq |x - y| + |y - 0| < \epsilon + |y|$$

so that

$$\epsilon < |y| = |f(y)| = |f(y) - 0| = |f(y) - f(x)|$$

Case $x \notin \mathbb{Q}$. Then $f(x) = x$, and there is clearly a rational y close enough to x that $|y - x| < \delta$. We then also have $f(y) = 0$ so that

$$|f(y) - f(x)| = |0 - x| = |x| = 2\epsilon > \epsilon$$

since $\epsilon > 0$. Hence in either case there is a y such that $|y - x| < \delta$ but $|f(y) - f(x)| \geq \epsilon$. This suffices to show that f is not continuous at x.

Problem 7

Let $f : A \to B$ and $g : C \to D$ be continuous functions. Let us define a map $f \times g : A \times C \to B \times D$ by the equation

$$(f \times g)(a, c) = (f(a), g(c)).$$

Show that $f \times g$ is continuous.

Solution: Consider any $(x, y) \in A \times C$ and any neighborhood V of $(f \times g)(x, y)$ in $B \times D$. Since V is open in $B \times D$, there is a basis element $U_B \times U_D$ of the product topology that contains $(f \times g)(x \times y)$ where $U_B \times U_D \subset V$. Then U_B and U_D are open in B and D, respectively. Since f is continuous, we then have that $U_A = f^{-1}(U_B)$ is open in A. Likewise $U_C = g^{-1}(U_D)$ is open in C since g is continuous. Then the set $U = U_A \times U_C$ is a basis element of the product topology $A \times C$ and therefore open.

Since $U_B \times U_D$ contains $(f \times g)(x, y) = (f(x), g(y))$ we have that $f(x) \in U_B$ and $g(y) \in U_D$. From this it follows that $x \in f^{-1}(U_B) = U_A$ and $y \in g^{-1}(U_D) = U_C$. Therefore $(x, y) \in U_A \times U_C = U$ so that U is a neighborhood of (x, y) in $A \times C$. Now consider any $(w, z) \in (f \times g)(U)$ so that there is an $(x', y') \in U = U_A \times U_C$ where $(w, z) = (f \times g)(x', y') = (f(x'), g(y'))$. Hence $w = f(x')$ and $x' \in U_A = f^{-1}(U_B)$ so that $w = f(x') \in U_B$. Similarly $z = g(y')$ and $y' \in U_C = g^{-1}(U_D)$ so that $z = g(y') \in U_D$. Thus $(w, z) \in U_B \times U_D$ so that also $(w, z) \in V$ since $U_B \times U_D \subset V$. This shows that $(f \times g)(U) \subset V$ since (w, z) was arbitrary. This suffices to show that $f \times g$ is continuous by Theorem 18.1 as desired.

Problem 8

Let $F : X \times Y \to Z$. We say that F is continuous in each variable separately if for each y_0 in Y, the map $h : X \to Z$ defined by $h(x) = F(x, y_0)$ is continuous, and for each $x_0 \in X$, the map $k : Y \to Z$ defined by $k(y) = F(x_0, y)$ is continuous. Show that if F is continuous, then F is continuous in each variable separately.

Solution: To show that F is continuous in x, consider any $y_0 \in Y$ and define $h : X \to Z$ by $h(x) = F(x, y_0)$. Now consider any $x \in X$ and any neighborhood V of $h(x) = F(x, y_0)$. Then V is an open set containing $h(x) = F(x, y_0)$ so that it is a neighborhood of $F(x, y_0)$. Since F is continuous, this means that there is neighborhood U' of (x, y_0) in $X \times Y$ such that $F(U') \subset V$ by Theorem 18.1. It then follows that there is a basis element $U_X \times U_Y$ of $X \times Y$ containing (x, y_0) where $U_X \times U_Y \subset U'$. Since $X \times Y$ is a product topology, we have that U_X is open in X and U_Y is open in Y. Then, since $(x, y_0) \in U_X \times U_Y$ we have that $x \in U_X$ and $y_0 \in U_Y$ so that U_X is a neighborhood of x in X.

So consider any $z \in h(U_X)$ so that $z = h(x')$ for some $x' \in U_X$. Then $(x', y_0) \in U_X \times U_Y$ so that also (x', y_0) in U' since $U_X \times U_Y \subset U'$. It then also follows that $z = h(x') = F(x', y_0) \in F(U')$ so that $z \in V$ since $F(U') \subset V$. This shows that $h(U_X) \subset V$ since z was arbitrary. It then follows that h is continuous by Theorem 18.1.

The proof that F is continuous in y is directly analogous.

Problem 9

Let $A \subset X$; let $f : A \to Y$ be continuous; let Y be Hausdorff. Show that if f may be extended to a continuous function $g : \bar{A} \to Y$, then g is uniquely determined by f.

Solution: Suppose that g_1 and g_2 are both continuous functions from $\bar{A}$ to Y that extend f so that $g_1(x) = g_2(x) = f(x)$ for all $x \in A$. Clearly $g_1 = g_2$ if and only if $g_1(x) = g_2(x)$ for all $x \in \bar{A}$. So suppose that this is not the case so that there is an $x_0 \in \bar{A}$ where $g_1(x_0) \neq g_2(x_0)$. Since Y is a Hausdorff space and $g_1(x_0)$ and $g_2(x_0)$ are distinct, there are disjoint neighborhoods V_1 and V_2 of $g_1(x_0)$ and $g_2(x_0)$, respectively. Then there are also neighborhoods U_1 and U_2 of x_0 such that $g_1(U_1) \subset V_1$ and $g_2(U_2) \subset V_2$ by Theorem 18.1 since both g_1 and g_2 are continuous. Now let $U = U_1 \cap U_2$ so that U is also a neighborhood of x_0. Since $x_0 \in \bar{A}$, it follows that U intersects A so that there is a $y \in U$ where also $y \in A$ by Theorem 17.5. Since $y \in A$ we have that $g_1(y) = g_2(y) = f(y)$. We also have that $y \in U_1$ and $y \in U_2$ since $U = U_1 \cap U_2$. Thus $g_1(y) \in g_1(U_1)$ so that $f(y) = g_1(y) \in V_1$ since $g_1(U_1) \subset V_1$. Similarly $f(y) = g_2(y) \in V_2$, but then we have that $f(y) \in V_1 \cap V_2$, which contradicts the fact that V_1 and V_2 are disjoint! Hence it must be that $g_1 = g_2$, which shows uniqueness.